KLAC KLA Corporation — 个股深度研究 (专业中文版) v20261016-0756

> 数据截至 2026-10-16 07:56 SGT｜价格 $174.46｜市值 227.7 bn USD｜财报货币 USD｜非投资建议。事实=CONFIRMED/DATA；推断=INFERENCE；假设=ASSUMPTION/HYPOTHESIS。

1. 投资论点 Investment Thesis (HYPOTHESIS)

KLA 在检测量测几乎寡占，先进制程/HBM/先进封装复杂度提升使检测强度上升；高 ROIC、强回购。

- **多头驱动**: WFE grows with N2/A16 and HBM; packaging inspection grows｜WFE 随 N2 与 HBM 增长；封装检测成长
- **空头驱动**: China revenue normalization, WFE digestion｜中国收入回落、WFE 消化
- **证伪条件 Falsifiers**: WFE outlook cut; process-control share loss｜WFE 展望下修；份额流失

2. 产品、客户与收入模式

- 主题 Theme: Process control / inspection 量测检测
- 业务分部 Segments: Semiconductor process control 半导体过程控制; Specialty semi process 特种工艺；PCB & component inspection PCB 检测
- 公司描述 (DATA, Yahoo/公司申报): KLA Corporation, together with its subsidiaries, provides process control and process-enabling solutions for manufacturing wafers, reticles/masks, chemicals/materials, integrated circuits (ICs), packaged ICs, and printed circuit boards. It operates through three segments: Semiconductor Process Control; Specialty Semiconductor Process; and PCB and Component Inspection. The company offers inspection and review tools to identify, locate, characterize, review, and analyze defects on various surfaces of patterned and unpatterned wafers; chemical process control equipment; wired and wireless sensor wafers and reticles; wafer defect inspection, review, and metrology systems; reticle inspection and metrology systems; wafer inspection and metrology systems; semiconductor software solutions; KLA Pro systems; and specialty semiconductor manufacturing, benchtop metrology, surface characterization, m

- 主要客户: TSMC, Samsung, Intel, SK hynix, Micron, Chinese fabs｜台积电、三星、英特尔、海力士、美光、中国晶圆厂

3. 行业、竞争、监管与供应链 (ANALYST FRAMEWORK)

- 供应商/上游: Optics, lasers, precision components｜光学、激光、精密部件
- 竞争者 (含业外): AMAT (e-beam), Hitachi High-Tech, Lasertec, Onto, Nova｜应材、日立高新、Lasertec、Onto、Nova
- 瓶颈/监管: China export controls, leading-edge capex cycle｜对华出口管制、先进制程 Capex 周期
- 关键 KPI: Revenue, GM%, China mix, backlog, buybacks｜收入、毛利率、中国占比、积压、回购
- 零部件现货/合约价: 本版未接入付费价格源(TrendForce/DRAMeXchange 等)——标注 **UNAVAILABLE**，在 KPI 监控中以定性方式跟踪。

4. 历史财务与现金流 (CONFIRMED, 季度, USD)

季度	收入	毛利	营业利润	净利润	FCF	营业利润率
2026-06-30	3.7 bn USD	2.2 bn USD	1.6 bn USD	1.4 bn USD	0.8 bn USD	42.5%
2026-03-31	3.4 bn USD	2.1 bn USD	1.4 bn USD	1.2 bn USD	0.6 bn USD	41.2%
2025-12-31	3.3 bn USD	2.0 bn USD	1.4 bn USD	1.1 bn USD	1.3 bn USD	41.3%
2025-09-30	3.2 bn USD	2.0 bn USD	1.3 bn USD	1.1 bn USD	1.1 bn USD	41.7%
2025-06-30	3.2 bn USD	2.0 bn USD	1.4 bn USD	1.2 bn USD	1.1 bn USD	42.6%
2025-03-31	N/A	N/A	N/A	N/A	N/A	N/A

资产负债: 现金 4.9 bn USD；总负债 6.2 bn USD；净现金 -1.2 bn USD。TTM FCF 3.8 bn USD（FCF 率 27.7%）。

5. 最新财报: 实际 vs 一致预期 vs 指引

季度	EPS 实际	EPS 一致预期	超预期%
2025-09-30	0.881	0.86172	0.0224
2025-12-31	0.885	0.87978	0.0058999998

季度	EPS 实际	EPS 一致预期	超预期%
2026-03-31	0.94	0.91703	0.025
2026-06-30	1.05	0.99866	0.0514

- 管理层指引 Guidance: 自动数据源不含指引——**UNAVAILABLE（待从公司 IR 新闻稿录入）**。
- 下次财报日 (Yahoo, TENTATIVE 直至公司确认): **2026-10-29**；本季 EPS 一致预期 1.17822（22 位分析师）；收入一致预期 4.0 bn USD。

6. 未来 2/4/6/8 季度驱动与价格合理性

期限	情景	期限价格现值	相对现价
2Q	Bear	$43.78	-74.9%
2Q	Base	$89.52	-48.7%
2Q	Bull	$141.62	-18.8%
4Q	Bear	$42.43	-75.7%
4Q	Base	$92.88	-46.8%
4Q	Bull	$152.94	-12.3%
6Q	Bear	$41.11	-76.4%
6Q	Base	$96.36	-44.8%
6Q	Bull	$165.06	-5.4%
8Q	Bear	$39.82	-77.2%
8Q	Base	$99.96	-42.7%
8Q	Bull	$178.04	2.1%

现价隐含的 EV/NTM-FCF (Base 路径):	期限
2Q	45.7x
4Q	41.6x
6Q	37.9x
8Q	34.5x

解读: 如果 8 季度后 Base 情景所需倍数仍远高于 25x，说明价格依赖更远期的增长。

7. 三情景 Bear / Base / Bull (HYPOTHESIS)

情景	Y1 增速	长期 FCF 率	WACC	永续 g	概率	每股价值	相对现价
Bear	9.5%	24.0%	13.4%	2.5%	25.0%	$30.24	-82.7%
Base	18.9%	30.0%	12.4%	3.0%	50.0%	$61.20	-64.9%
Bull	24.6%	34.5%	11.9%	3.5%	25.0%	$96.51	-44.7%

概率加权价值 **$62.29**（相对现价 -64.3%）。

8. 估值: DCF / Reverse DCF / ROIC-WACC / 同业 / 敏感度

- DCF Base: EV 81.1 bn USD；终值占比 47.6%。
- **Reverse DCF 反向 DCF**: 在 Base 利润率与 WACC 下，现价隐含 Y1 收入增速 ≈ **48.1%**；一致预期 FY+1 增速 18.9%。
- WACC (CAPM) = rf 5.2% + β 1.45 × ERP 5.0% = 12.4%。
- SOTP segments 分部: Semiconductor process control 半导体过程控制; Specialty semi process 特种工艺; PCB & component inspection PCB 检测. **Status: framework only** — segment revenues are INPUT REQUIRED from the latest 10-K/10-Q segment note (not auto-fed, left blank to avoid fabrication)｜分部收入需按最新财报手动录入，未自动填充以避免编造。
- 同业比较 Comps (DATA): | Ticker | 市值 | EV/Sales | Fwd P/E | 收入增速 | 毛利率 | FCF 收益率 | |—|—|—|—|—|—|—| | KLAC | 227.7 bn USD | 16.9x | 25.9x | 15.2% | 61.3% | 1.7% | | AMAT | 341.3 bn USD | 11.0x | 23.3x | 24.8% | 49.4% | 0.9% | | LRCX | 347.2 bn USD | 14.9x | 23.6x | 30.0% | 50.5% | 0.9% | | ASML | 636.6 bn USD | 996.5x | 26.7x | 21.3% | 52.7% | 1.3% | | 8035.T | 4,705.4 bn USD | 8.8x | 8.1x | 33.3% | 45.6% | 6.3% |
- 敏感度: 见 Excel `Sensitivity` 表 (WACC × 永续增长率)。

9. 风险、反方论证与证伪

- 反方观点 Counter-thesis: China revenue normalization, WFE digestion｜中国收入回落、WFE 消化
- 证伪指标: WFE outlook cut; process-control share loss｜WFE 展望下修；份额流失
- 模型风险: 长期利润率与 WACC 为假设；终值占比高时估值对假设极敏感。

10. Source Ledger 来源账本

Item	Value	Source/time
Price 股价	174.5	Yahoo Finance quote, fetched 2026-10-16T07:56 SGT
Market cap 市值	2.277e+11	Yahoo info.marketCap 2026-10-16T07:56
Cash 现金	4.902e+09	Yahoo info.totalCash (latest filing) 2026-10-16T07:56
Debt 负债	6.152e+09	Yahoo info.totalDebt (latest filing) 2026-10-16T07:56
Revenue TTM 收入	1.358e+10	Yahoo info.totalRevenue (TTM, USD) 2026-10-16T07:56
FCF TTM 自由现金流	3.767e+09	Sum of last 4 quarterly FCF (q_cash); quarters ['2026-06-30', '2026-03-31', '2025-12-31', '2025-09-30'] (USD)
Beta	1.447	Yahoo info.beta (5Y monthly)
Risk-free 10Y 无风险利率	0.0519	FRED DGS10 2026-10-14
FX USD per USD	1	n/a (USD)
Consensus rev growth FY0	0.336	Yahoo revenue_estimate 0y
Consensus rev growth FY+1	0.1892	Yahoo revenue_estimate +1y

11. Assumption Register 假设登记表

Assumption	Value	Rationale
ERP 股权风险溢价	0.05	ASSUMPTION: 5.0% house mature-market ERP
Beta clamp	[0.8,1.6] → 1.45	ASSUMPTION: limit extreme betas
Tax rate (ROIC)	0.18	ASSUMPTION: blended cash tax 18%

Assumption	Value	Rationale
Year-1 growth (Base)	0.1892	Consensus FY+1 revenue growth (Yahoo revenue_estimate '+1y'); capped at 100%
Growth fade	linear Y1→terminal by Y10	ASSUMPTION
Starting FCF margin	0.2774	DATA: FCF TTM / Revenue TTM, floored -50%
Long-run FCF margin (Base)	0.3	ASSUMPTION (analyst framework, reached by Y5; see profile)
Bear/Bull Y1 growth multiplier	0.5x / 1.3x	ASSUMPTION
Bear/Bull margin multiplier	0.8x / 1.15x	ASSUMPTION
WACC adj Bear/Bull	+1.0pp / -0.5pp	ASSUMPTION
Terminal g Bear/Base/Bull	2.5% / 3.0% / 3.5%	ASSUMPTION
Scenario probabilities	25% / 50% / 25%	ASSUMPTION
Horizon exit EV/NTM-FCF Bear/Base/Bull	15x / 25x / 35x	ASSUMPTION
Per-share conversion	value × fx ÷ market cap × price	METHOD: avoids ADR/share-class mismatch

12. Model Provenance 模型溯源

- Model file: `KLAC_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions)
- Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx)
- Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT
- Version: v20261016-0756. Prior versions are never overwritten (see company history page).

13. 版本与事件时间轴

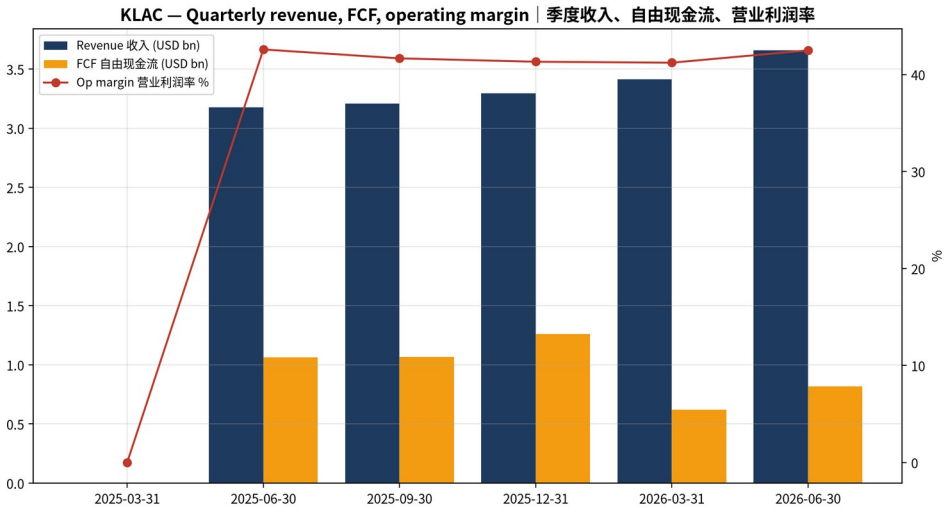
- v20261016-0756: 首版生产系统报告 (27 家同步)。下一事件: 财报 2026-10-29 (TENTATIVE)。

图片报告 Image Report



1_price_vs_value

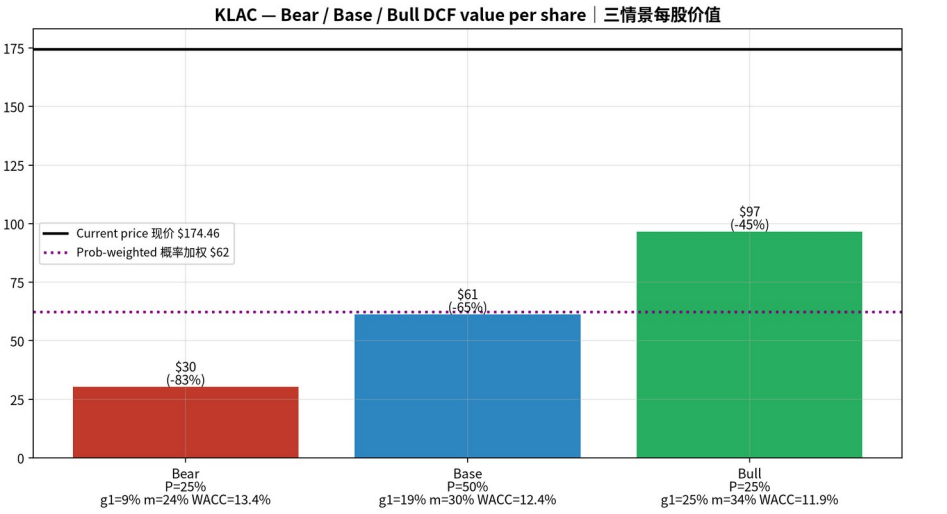
股价vs 三情景价值Price vs scenario values



Claim class 声明类别: CONFIRMED (reported) | Source 来源: Company filings via Yahoo Finance quarterly statements
Thinking 思考: Is FCF growing with revenue? Divergence = capex/working-capital intensity → check cash-flow statement. | Not investment advice 非投资建议

2_financials

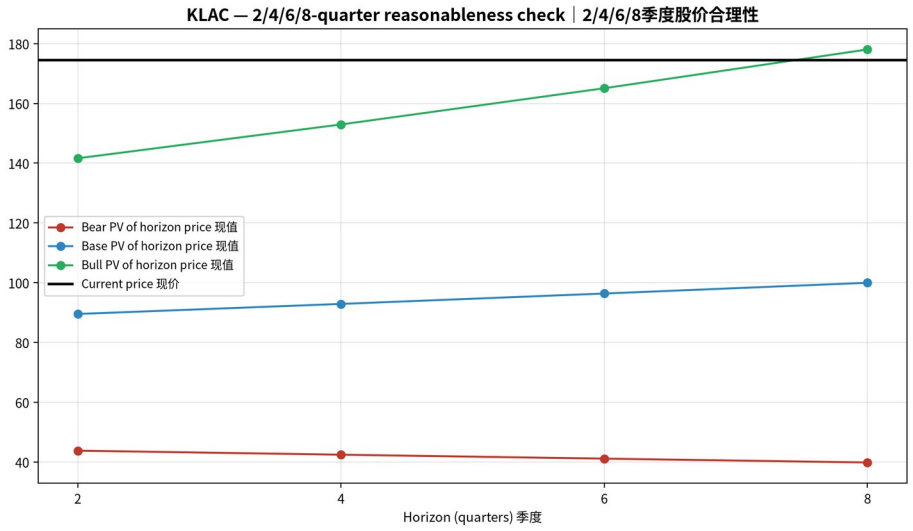
季度财务 Quarterly financials



Claim class 声明类别: HYPOTHESIS (model output) | Source 来源: Model (live formulas) — inputs in Sources/Assumptions sheets
Thinking 思考: Which assumption moves value most? Check Sensitivity sheet; scenario width = uncertainty. | Not investment advice 非投资建议

3_scenarios

三情景 Scenarios



Claim class 声明类别: HYPOTHESIS | Source 来源: Model Horizons sheet; exit EV/NTM-FCF 15x/25x/35x (ASSUMPTION)
Thinking 思考: If even Bull PV < price at 8Q, price requires assumptions beyond Bull within 2 years. | Not investment advice 非投资建议

4_horizons

2/4/6/8 季度检验Horizon check



Reverse DCF 反向DCF: price implies Y1 growth ≈ 48%

Consensus FY+1 growth 一致预期: 19% → gap = expectation the price embeds 价格隐含预期差

Claim class 声明类别: INFERENCE / HYPOTHESIS | Source 来源: Profile (analyst framework) + model
Thinking 思考: Read left→right: each box must have evidence; the falsifier box tells you what would prove the thesis wrong. | Not investment advice 非投资建议

5_logic_map

研究逻辑图Logic map

KLAC KLA Corporation — Equity Deep Dive (Professional English) v20261016-0756

Data as of 2026-10-16 07:56 SGT | Price $174.46 | Mkt cap 227.7 bn USD | Reporting currency USD | Not investment advice.

1. Investment thesis (HYPOTHESIS)

KLA is a near-oligopoly in inspection/metrology; rising complexity at leading-edge, HBM and advanced packaging increases process-control intensity. High ROIC, strong buybacks.

- **Bull drivers**: WFE grows with N2/A16 and HBM; packaging inspection grows ｜WFE 随 N2 与 HBM 增长；封装检测成长
- **Bear drivers**: China revenue normalization, WFE digestion｜中国收入回落、WFE 消化
- **Falsifiers**: WFE outlook cut; process-control share loss｜WFE 展望下修；份额流失

2. Products, customers, revenue model

- Theme: Process control / inspection 量测检测 | Segments: Semiconductor process control 半导体过程控制; Specialty semi process 特种工艺; PCB & component inspection PCB 检测
- Business description (DATA): KLA Corporation, together with its subsidiaries, provides process control and process-enabling solutions for manufacturing wafers, reticles/masks, chemicals/materials, integrated circuits (ICs), packaged ICs, and printed circuit boards. It operates through three segments: Semiconductor Process Control; Specialty Semiconductor Process; and PCB and Component Inspection. The company offers inspection and review tools to identify, locate, characterize, review, and analyze defects on various surfaces of patterned and unpatterned wafers; chemical process control equipment; wired and wireless sensor wafers and reticles; wafer defect inspection, review, and metrology systems; reticle inspection and metrology systems; wafer inspection and metrology systems; semiconductor software solutions; KLA Pro systems; and specialty semiconductor manufacturing, benchtop metrology, surface characterization, m
- Customers: TSMC, Samsung, Intel, SK hynix, Micron, Chinese fabs｜台积电、三星、英特尔、海力士、美光、中国晶圆厂

3. Industry, competition, regulation, supply chain (ANALYST FRAMEWORK)

- Suppliers: Optics, lasers, precision components｜光学、激光、精密部件
- Competitors: AMAT (e-beam), Hitachi High-Tech, Lasertec, Onto, Nova｜应材、日立高新、Lasertec、Onto、Nova
- Bottlenecks/regulation: China export controls, leading-edge capex cycle｜对华出口管制、先进制程 Capex 周期
- KPIs: Revenue, GM%, China mix, backlog, buybacks｜收入、毛利率、中国占比、积压、回购
- Component spot/contract prices: paid feeds not connected — **UNAVAILABLE**, tracked qualitatively.

4. Historical financials & cash flow (CONFIRMED, quarterly, USD)

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
2026-06-30	3.7 bn USD	2.2 bn USD	1.6 bn USD	1.4 bn USD	0.8 bn USD	42.5%
2026-03-31	3.4 bn USD	2.1 bn USD	1.4 bn USD	1.2 bn USD	0.6 bn USD	41.2%
2025-12-31	3.3 bn USD	2.0 bn USD	1.4 bn USD	1.1 bn USD	1.3 bn USD	41.3%
2025-09-30	3.2 bn USD	2.0 bn USD	1.3 bn USD	1.1 bn USD	1.1 bn USD	41.7%
2025-06-30	3.2 bn USD	2.0 bn USD	1.4 bn USD	1.2 bn USD	1.1 bn USD	42.6%
2025-03-31	N/A	N/A	N/A	N/A	N/A	N/A

Cash 4.9 bn USD; debt 6.2 bn USD; TTM FCF 3.8 bn USD (FCF margin 27.7%).

5. Latest earnings: actual vs consensus vs guidance

Quarter	EPS actual	EPS consensus	Surprise
2025-09-30	0.881	0.86172	0.0224
2025-12-31	0.885	0.87978	0.0058999998
2026-03-31	0.94	0.91703	0.025
2026-06-30	1.05	0.99866	0.0514

- Management guidance: not in automated feed — **UNAVAILABLE (to be entered from IR release)**.
- Next earnings (Yahoo; TENTATIVE until company-confirmed): **2026-10-29**; current-quarter EPS consensus 1.17822 (22 analysts); revenue consensus 4.0 bn USD.

6. 2/4/6/8-quarter drivers & price reasonableness

Horizon	Scenario	PV of horizon price	vs current
2Q	Bear	$43.78	-74.9%
2Q	Base	$89.52	-48.7%
2Q	Bull	$141.62	-18.8%
4Q	Bear	$42.43	-75.7%
4Q	Base	$92.88	-46.8%
4Q	Bull	$152.94	-12.3%
6Q	Bear	$41.11	-76.4%
6Q	Base	$96.36	-44.8%
6Q	Bull	$165.06	-5.4%
8Q	Bear	$39.82	-77.2%
8Q	Base	$99.96	-42.7%
8Q	Bull	$178.04	2.1%

Current-price implied EV/NTM-FCF on Base path:	Horizon
2Q	45.7x
4Q	41.6x
6Q	37.9x
8Q	34.5x

7. Bear / Base / Bull (HYPOTHESIS)

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Bear	9.5%	24.0%	13.4%	2.5%	25.0%	$30.24	-82.7%
Base	18.9%	30.0%	12.4%	3.0%	50.0%	$61.20	-64.9%
Bull	24.6%	34.5%	11.9%	3.5%	25.0%	$96.51	-44.7%

Probability-weighted value **$62.29** (-64.3% vs price).

8. Valuation: DCF / Reverse DCF / ROIC-WACC / comps / sensitivity

- Base DCF EV 81.1 bn USD; terminal share 47.6%.
- **Reverse DCF**: at Base margin/WACC the price implies Y1 revenue growth ≈ **48.1%** vs consensus FY+1 18.9%.
- WACC (CAPM) = rf 5.2% + β 1.45 × ERP 5.0% = 12.4%.
- SOTP segments: Semiconductor process control 半导体过程控制; Specialty semi process 特种工艺; PCB & component inspection PCB 检测 — framework only; segment revenue inputs pending from filings.
- Comps (DATA): see table in Chinese section / Excel `Comps`.
- Sensitivity: Excel `Sensitivity` (WACC × terminal g).

9. Risks, counterarguments, falsifiers

China revenue normalization, WFE digestion｜中国收入回落、WFE 消化 | Falsifiers: WFE outlook cut; process-control share loss｜WFE 展望下修；份额流失

10–12. Source Ledger / Assumption Register / Model Provenance

See tables above (shared, bilingual). - Model file: `KLAC_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions) - Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx) - Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT - Version: v20261016-0756. Prior versions are never overwritten (see company history page).